

5-Mark Questions & Answers -- MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Explain the Von Neumann architecture in detail with its advantages and limitations.

1. Detailed Concept

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2. Key Advantages

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3. Limitations (Von Neumann Bottleneck)

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Q2. Explain all addressing modes of the 8086 microprocessor with examples.

1. Immediate Mode: Operand is a constant value inside instruction.

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Q3. Discuss the performance measures of computer architecture (CPU time, clock cycles, MIPS, benchmarks) in detail.

1. CPU Execution Time Formula

CPU Ti

CPU Ti

2. Clock Cycles & CPI

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3. MIPS (Million Instructions Per Second)

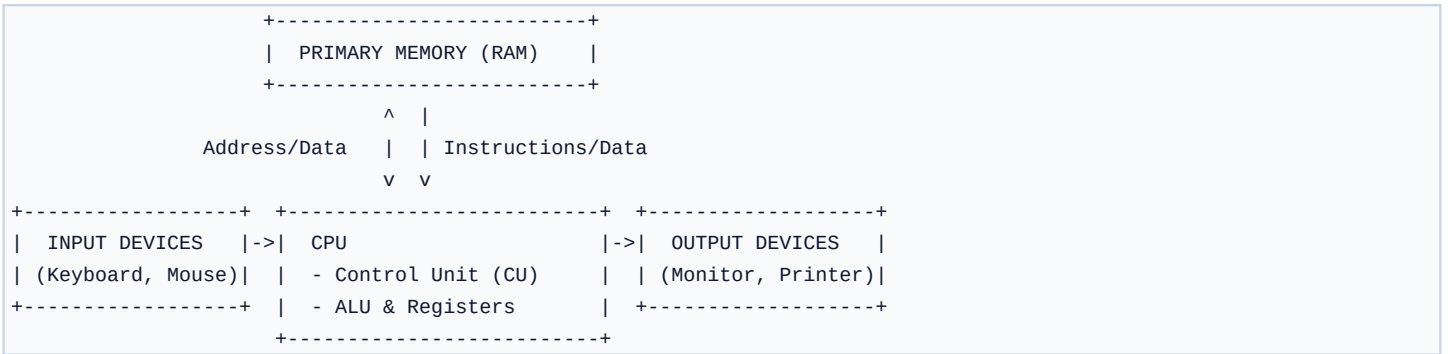
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4. Benchmarks

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Q4. Explain the basic organization and block-level description of the functional units of a computer.



1. Input Unit: Accepts external user data/programs and converts them into binary code.

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Q5. Compare Harvard architecture with Von Neumann architecture.

Parameter	Von Neumann Architecture	Harvard Architecture
Memory System	Shared single memory for instructions	Separate physical memory blocks for in
Bus System	Shared single bus for instruction fetch	Separate instruction bus and data bus.
Pipelining Efficiency	Lower efficiency due to bus conflict b	Higher efficiency (simultaneous instru
Complexity & Cost	Simple structure, lower hardware cost.	Complex memory controller, higher cost
Applications	General-purpose desktop computers, ser	DSP processors, Microcontrollers (ARM